Guiding-center collision FP coefficients in an isotropic non-uniform field-particle distribution

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1 First-order corrections and gyroaveraging of the particle kinetic momentum

From Brizard[1] we have the following first-order expression for the particle gyroradius, calculated using the Lie-transform generating vectors,

$$\boldsymbol{\rho}_\epsilon = \boldsymbol{\rho}_0 + \epsilon\boldsymbol{\rho}_1 = \boldsymbol{\rho}_0 - \alpha\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi} - \beta\boldsymbol{\rho}_0 - \gamma\widehat{\mathsf{b}} \tag{1}$$

where $\boldsymbol{\rho}_0 \equiv \widehat{\mathsf{b}} \times \mathbf{p}/m\Omega$ and

$$\alpha = \frac{v_\parallel}{\Omega}\mathbb{A}_2 \colon \boldsymbol{\nabla}\widehat{\mathsf{b}} + \frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\cdot\left(\boldsymbol{\nabla}\ln B + \frac{mv_\parallel^2}{2\mu B}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right) \tag{2}$$

$$\beta = \boldsymbol{\rho}_0\cdot\left(\frac{1}{2}\boldsymbol{\nabla}\ln B + \frac{mv_\parallel^2}{2\mu B}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right) + \frac{v_\parallel}{2\Omega}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}} - \mathbb{A}_1\colon\boldsymbol{\nabla}\widehat{\mathsf{b}}\right) \tag{3}$$

$$\gamma = 2\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\cdot\left(\frac{v_\parallel}{\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right) - \frac{\mu B}{m\Omega^2}\left(\frac{1}{2}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\ln B + \mathbb{A}_2\colon\boldsymbol{\nabla}\widehat{\mathsf{b}}\right) \tag{4}$$

The tensors $\mathbb{A}_1 \equiv \frac{1}{2}\left(\widehat{\rho}\widehat{\perp} + \widehat{\perp}\widehat{\rho}\right)$ and $\mathbb{A}_2$ are related by $\mathbb{A}_1 = \partial\mathbb{A}_2/\partial\varphi$. We then use the following expressions,

$$\mathbf{p}_\epsilon \equiv m\frac{d_\epsilon\mathbf{X}_\epsilon}{dt} = m\left[\mathbf{v}_{gc} + \left(\Omega\frac{\partial}{\partial\varphi} + \mathbf{v}_{gc}\cdot\boldsymbol{\nabla}^*\right)\boldsymbol{\rho}_\epsilon\right] \tag{5}$$

$$\mathbf{v}_{gc} = v_\parallel\widehat{\mathsf{b}} + \frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*}\times\left(\mu\boldsymbol{\nabla}B + mv_\parallel^2\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right) \equiv v_\parallel\widehat{\mathsf{b}} + \epsilon\mathbf{v}_B \tag{6}$$

$$\boldsymbol{\nabla}^* \equiv \boldsymbol{\nabla} + \left(\mathbf{R} + \frac{1}{2}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right)\widehat{\mathsf{b}}\right)\frac{\partial}{\partial\varphi} \tag{7}$$

$$\boldsymbol{\nabla}\boldsymbol{\rho}_0 = -\frac{1}{2}\boldsymbol{\nabla}\ln B\;\boldsymbol{\rho}_0 - \left(\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\boldsymbol{\rho}_0\right)\;\widehat{\mathsf{b}} - \mathbf{R}\;\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi} \tag{8}$$

$$\mathbf{p} = v_\parallel\widehat{\mathsf{b}} + m\Omega\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi} \tag{9}$$

in order to get the first-order particle momentum:

$$
\begin{aligned}
\mathbf{p}_\epsilon \equiv \mathbf{p}+\epsilon\mathbf{p}_1+\mathcal{O}\left(\epsilon^2\right) =& m\left[\mathbf{v}_{gc}+\Omega\frac{\partial\boldsymbol{\rho}_\epsilon}{\partial\varphi}+\mathbf{v}_{gc}\cdot\boldsymbol{\nabla}\boldsymbol{\rho}_0+\mathbf{v}_{gc}\cdot\left(\mathbf{R}+\frac{1}{2}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right)\widehat{\mathsf{b}}\right)\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right]\\
=&\mathbf{p}+\epsilon\left[\frac{\widehat{\mathsf{b}}}{\Omega}\times\left(\mu\boldsymbol{\nabla}B+mv_\parallel^2\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right)\right.\\
&-m\Omega\left(\frac{\partial\alpha}{\partial\varphi}+\beta\right)\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}-m\Omega\left(\alpha-\frac{\partial\beta}{\partial\varphi}\right)\boldsymbol{\rho}_0-m\Omega\frac{\partial\gamma}{\partial\varphi}\widehat{\mathsf{b}}\\
&+mv_\parallel\widehat{\mathsf{b}}\cdot\left(-\frac{1}{2}\boldsymbol{\nabla}\ln B\,\boldsymbol{\rho}_0-\left(\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\boldsymbol{\rho}_0\right)\,\widehat{\mathsf{b}}-\mathbf{R}\,\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right)\\
&\left.+mv_\parallel\widehat{\mathsf{b}}\cdot\left(\mathbf{R}+\frac{1}{2}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right)\widehat{\mathsf{b}}\right)\,\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right]+\mathcal{O}\left(\epsilon^2\right)\\
=&\mathbf{p}+\epsilon\left[\frac{\widehat{\mathsf{b}}}{\Omega}\times\left(\mu\boldsymbol{\nabla}B+mv_\parallel^2\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right)\right.\\
&-m\Omega\left(\frac{\partial\gamma}{\partial\varphi}+\frac{v_\parallel}{\Omega}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\boldsymbol{\rho}_0\right)\right)\widehat{\mathsf{b}}\\
&-m\Omega\left(\beta+\frac{\partial\alpha}{\partial\varphi}-\frac{v_\parallel}{2\Omega}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right)\right)\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\\
&\left.+m\Omega\left(\alpha-\frac{\partial\beta}{\partial\varphi}-\frac{v_\parallel}{2\Omega}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\ln B\right)\right)\boldsymbol{\rho}_0\right]+\mathcal{O}\left(\epsilon^2\right)
\end{aligned}
\tag{10}
$$

We can readily see that

$$
\begin{aligned}
|\mathbf{p}_\epsilon|^2=|\mathbf{p}|^2+2\epsilon\mathbf{p}\cdot\mathbf{p}_1=&|\mathbf{p}|^2+2\epsilon\left[\frac{1}{\Omega}\left(\mu\boldsymbol{\nabla}B+mv_\parallel^2\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right)\cdot\left(\mathbf{p}\times\widehat{\mathsf{b}}\right)\right.\\
&-mv_\parallel\Omega\left(\frac{\partial\gamma}{\partial\varphi}+\frac{v_\parallel}{\Omega}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\boldsymbol{\rho}_0\right)\right)\\
&\left.-m^2\Omega^2\left(\beta+\frac{\partial\alpha}{\partial\varphi}-\frac{v_\parallel}{2\Omega}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right)\right)\left|\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right|^2\right]+\mathcal{O}\left(\epsilon^2\right)
\end{aligned}
$$

Using $\left|\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right|^2=|\boldsymbol{\rho}_0|^2=2\mu B/m\Omega^2$ and inserting the expressions for α,β and γ leads to

$$
|\mathbf{p}_\epsilon|^2=|\mathbf{p}|^2+\mathcal{O}\left(\epsilon^2\right)=2m\mathcal{E}+\mathcal{O}\left(\epsilon^2\right) \tag{11}
$$

Gyro-averaging of (10) gives

$$
\begin{aligned}
\langle\mathbf{p}_\epsilon\rangle =&m\mathbf{v}_{gc}-m\Omega\left(\left\langle\frac{\partial\gamma}{\partial\varphi}\right\rangle+\frac{v_\parallel}{\Omega}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\langle\boldsymbol{\rho}_0\rangle\right)\right)\widehat{\mathsf{b}}\\
&-m\Omega\left\langle\left(\beta+\frac{\partial\alpha}{\partial\varphi}\right)\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle-\frac{mv_\parallel}{2}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right)\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle\\
&+m\Omega\left\langle\left(\alpha-\frac{\partial\beta}{\partial\varphi}\right)\boldsymbol{\rho}_0\right\rangle-\frac{mv_\parallel}{2}\left(\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\ln B\right)\langle\boldsymbol{\rho}_0\rangle+\mathcal{O}\left(\epsilon^2\right)
\end{aligned}
$$

Gyro-averages of φ-derivatives are obviously zero. Using $\langle\boldsymbol{\rho}_0\rangle=0$, and performing an integration by parts gives:

$$
\begin{aligned}
\langle\mathbf{p}_\epsilon\rangle =&m\mathbf{v}_{gc}+m\Omega\left\langle\left(\alpha+\frac{\partial^2\alpha}{\partial\varphi^2}\right)\boldsymbol{\rho}_0\right\rangle+\mathcal{O}\left(\epsilon^2\right)\\
=&m\mathbf{v}_{gc}+mv_\parallel\left\langle\left(\mathbb{A}_2\colon\boldsymbol{\nabla}\widehat{\mathsf{b}}+\frac{\partial\mathbb{A}_1}{\partial\varphi}\colon\boldsymbol{\nabla}\widehat{\mathsf{b}}\right)\boldsymbol{\rho}_0\right\rangle+\mathcal{O}\left(\epsilon^2\right)
\end{aligned}
$$

Since $\left\langle\left(\partial\mathbb{A}_1/\partial\varphi\colon\boldsymbol{\nabla}\widehat{\mathsf{b}}\right)\boldsymbol{\rho}_0\right\rangle=-\left\langle\left(\mathbb{A}_1\colon\boldsymbol{\nabla}\widehat{\mathsf{b}}\right)\partial\boldsymbol{\rho}_0/\partial\varphi\right\rangle=-\left\langle\left(\mathbb{A}_2\colon\boldsymbol{\nabla}\widehat{\mathsf{b}}\right)\boldsymbol{\rho}_0\right\rangle$ when integrating by parts (due to the fact that $\partial^2\boldsymbol{\rho}_0/\partial\varphi^2=-\boldsymbol{\rho}_0$), we readily have

$$
\langle\mathbf{p}_\epsilon\rangle=m\mathbf{v}_{gc}+\mathcal{O}\left(\epsilon^2\right) \tag{12}
$$

2 Calculation of guiding-center coefficients

The drag vector and diffusion tensor have the following expression in an isotropic field-particle distribution:

$$\mathbf{K} = -\nu\left(\mathbf{x}, p\right)\mathbf{p}$$

$$\mathbb{D} = D_l\left(\mathbf{x}, p\right)\mathbf{pp}/p^2 + D_t\left(\mathbf{x}, p\right)\left(\mathbb{I} - \mathbf{pp}/p^2\right)$$

Note that in practice the coefficients' spatial dependance will essentially be in ψ. The guiding-center transformation yields the following expressions

$$\mathbf{K}_\epsilon \equiv \mathsf{T}_\epsilon^{-1}\mathbf{K} = \nu_\epsilon\left(\mathbf{X}, p\right)\mathbf{p}_\epsilon$$

$$\mathbb{D}_\epsilon \equiv \mathsf{T}_\epsilon^{-1}\mathbb{D} = D_{l\epsilon}\left(\mathbf{X}, p\right)\mathbf{p}_\epsilon\mathbf{p}_\epsilon/p^2 + D_{t\epsilon}\left(\mathbf{X}, p\right)\left(\mathbb{I} - \mathbf{p}_\epsilon\mathbf{p}_\epsilon/p^2\right)$$

where the coefficients ν_ϵ, $D_{l\epsilon}$ and $D_{t\epsilon}$are determined at first-order in the density-temperature nonuniformity $\epsilon_{n,T}$ which we will assume is equivalent to the magnetic uniformity (i.e., $\epsilon_{n,T} \sim \epsilon_B \sim \epsilon$)

$$\nu_\epsilon = \nu - \epsilon\boldsymbol{\rho}_0 \cdot \boldsymbol{\nabla}\nu + \mathcal{O}\left(\epsilon^2\right) \tag{13}$$

$$D_{l\epsilon} = D_l - \epsilon\boldsymbol{\rho}_0 \cdot \boldsymbol{\nabla}D_l + \mathcal{O}\left(\epsilon^2\right) \tag{14}$$

$$D_{t\epsilon} = D_t - \epsilon\boldsymbol{\rho}_0 \cdot \boldsymbol{\nabla}D_t + \mathcal{O}\left(\epsilon^2\right) \tag{15}$$

The FP guiding-center coefficients are deduced by the following gyro-averages,

$$K_{gc}^\alpha = \left\langle \mathbf{K}_\epsilon \cdot \boldsymbol{\Delta}_{gc}^\alpha \right\rangle \tag{16}$$

$$D_{gc}^{\alpha\beta} = \left\langle \boldsymbol{\Delta}_{gc}^\alpha \cdot \mathbb{D}_\epsilon \cdot \boldsymbol{\Delta}_{gc}^\beta \right\rangle \tag{17}$$

where the projection vectors $\boldsymbol{\Delta}_{gc}^\alpha$ read

$$\boldsymbol{\Delta}_{gc}^{\mathbf{X}} = -\left[\frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \boldsymbol{\nabla}^*\mathbf{X}_\epsilon\right]^T - \frac{\partial\boldsymbol{\rho}_\epsilon}{\partial\mathcal{E}}\,\mathbf{v}_{gc} \tag{18}$$

$$\boldsymbol{\Delta}_{gc}^{\mathcal{E}} = \frac{\mathbf{p}_\epsilon}{m} \tag{19}$$

$$\boldsymbol{\Delta}_{gc}^{\mu} = \frac{\Omega}{B}\frac{\partial\boldsymbol{\rho}_\epsilon}{\partial\varphi} \tag{20}$$

where for given vector and tensor $\mathbf{a}$ and $\mathbb{B}$, $(\mathbf{a} \times \mathbb{B})_l^i = \epsilon^{ijk}a_jB_{kl} = -(\mathbb{B} \times \mathbf{a})_l^i$

2.1 Momentum-space guiding-center coefficients

2.1.1 Calculation of $K_{gc}^{\mathcal{E}}$

Using the expressions above and the properties of $\mathbf{p}_\epsilon$, we immediately have:

$$K_{gc}^{\mathcal{E}} \equiv \frac{1}{m}\left\langle \mathbf{K}_\epsilon \cdot \mathbf{p}_\epsilon \right\rangle = -2\mathcal{E}\left\langle \nu_\epsilon \right\rangle = -2\nu\mathcal{E} + \mathcal{O}\left(\epsilon^2\right) \tag{21}$$

2.1.2 Calculation of K_{gc}^{μ}

The expression for K_{gc}^{μ} reads

$$\begin{aligned}
K_{gc}^{\mu} \equiv& \frac{\Omega}{B}\left\langle \mathbf{K}_\epsilon \cdot \frac{\partial\boldsymbol{\rho}_\epsilon}{\partial\varphi} \right\rangle \\
=& -\frac{\Omega}{B}\left\langle \nu_\epsilon\mathbf{p}_\epsilon \cdot \frac{\partial\boldsymbol{\rho}_\epsilon}{\partial\varphi} \right\rangle \\
=& -\frac{\Omega}{B}\left[\nu\left\langle \mathbf{p} \cdot \frac{\partial\boldsymbol{\rho}_0}{\partial\varphi} \right\rangle + \nu\left\langle \mathbf{p} \cdot \frac{\partial\boldsymbol{\rho}_1}{\partial\varphi} \right\rangle + \nu\left\langle \mathbf{p}_1 \cdot \frac{\partial\boldsymbol{\rho}_0}{\partial\varphi} \right\rangle - \left\langle \mathbf{p} \cdot \frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\boldsymbol{\rho}_0 \right\rangle \cdot \boldsymbol{\nabla}\nu\right] + \mathcal{O}\left(\epsilon^2\right) \\
=& -2\mu\nu + \nu\frac{\Omega}{B}\left[m\Omega\left\langle \left(\alpha - \frac{\partial\beta}{\partial\varphi}\right)\boldsymbol{\rho}_0 \cdot \frac{\partial\boldsymbol{\rho}_0}{\partial\varphi} \right\rangle + \frac{2\mu B}{\Omega}\left\langle \frac{\partial\alpha}{\partial\varphi} + \beta \right\rangle + mv_\parallel\left\langle \frac{\partial\gamma}{\partial\varphi} \right\rangle\right] \\
& - \mu\nu\widehat{\mathsf{b}} \times \left(\boldsymbol{\nabla}\ln B + \frac{mv_\parallel^2}{\mu B}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla}\widehat{\mathsf{b}}\right) \cdot \left\langle \frac{\partial\boldsymbol{\rho}_0}{\partial\varphi} \right\rangle + 2\mu\nu\left[\left\langle \beta + \frac{\partial\alpha}{\partial\varphi} \right\rangle - \frac{v_\parallel}{2\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}}\right] \\
& - 2\mu\left\langle \boldsymbol{\rho}_0 \right\rangle \cdot \boldsymbol{\nabla}\nu + \mathcal{O}\left(\epsilon^2\right) \\
=& -2\mu\nu\left[1 - 2\left\langle \beta \right\rangle + \frac{v_\parallel}{2\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}}\right] + \mathcal{O}\left(\epsilon^2\right)
\end{aligned}$$

where we used $\boldsymbol{\rho}_0 \cdot \partial\boldsymbol{\rho}_0/\partial\varphi = 0$ and $|\partial\boldsymbol{\rho}_0/\partial\varphi|^2 = 2\mu B/m\Omega^2$. Inserting the expressions for β gives

$$K_{gc}^{\mu} = -\mu\nu\left(2 - \frac{v_{\parallel}}{\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right) + \mathcal{O}\left(\epsilon^2\right) \tag{22}$$

2.1.3 Calculation of $\mathbf{K}_{gc}^{\mathbf{X}}$

The expression for $\mathbf{K}_{gc}^{\mathbf{X}}$ reads

$$\begin{aligned}
\mathbf{K}_{gc}^{\mathbf{X}} \equiv& \left\langle \boldsymbol{\Delta}_{gc}^{\mathbf{X}T}\cdot\mathbf{K}_{\epsilon}\right\rangle \\
=& -\left\langle \nu_{\epsilon}\boldsymbol{\Delta}_{gc}^{\mathbf{X}T}\cdot\mathbf{p}_{\epsilon}\right\rangle \\
=& \nu\frac{\widehat{\mathsf{b}}}{m\Omega_{\parallel}^{*}}\times\left[\langle\mathbf{p}_{\epsilon}\rangle - \epsilon m v_{\parallel}\left(\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\langle\boldsymbol{\rho}_0\rangle\right)\right] - 2\nu\left(\frac{\mu B}{m\Omega^2}\langle\mathbb{A}_2\rangle : \boldsymbol{\nabla}\widehat{\mathsf{b}} + 2\frac{v_{\parallel}}{\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle\right)\widehat{\mathsf{b}} \\
&+ \boldsymbol{\nabla}\nu\cdot\left\langle\boldsymbol{\rho}_0\,\frac{\widehat{\mathsf{b}}}{m\Omega}\times\mathbf{p}\right\rangle \\
=& \nu\frac{\widehat{\mathsf{b}}}{m\Omega}\times\mathbf{v}_B + \frac{\mu B}{m\Omega^2}\boldsymbol{\nabla}\nu\cdot\left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}}\right) \\
=& -\epsilon^2\frac{\mu B}{m\Omega^2}\left[\nu\left(\boldsymbol{\nabla}\ln B + \frac{v_{\parallel}^2}{\Omega^2}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right) - \boldsymbol{\nabla}\nu\right]\cdot\left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}}\right) + \mathcal{O}\left(\epsilon^3\right)
\end{aligned}$$

where we used $\boldsymbol{\rho}_0 \equiv \widehat{\mathsf{b}}\times\mathbf{p}/m\Omega$ and $\langle\boldsymbol{\rho}_0\,\boldsymbol{\rho}_0\rangle = \left(\mu B/m\Omega^2\right)\left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}}\right)$

2.1.4 Calculation of $D_{gc}^{\mathcal{E}\mathcal{E}}$

Similarly to $K_{gc}^{\mathcal{E}}$, we have

$$D_{gc}^{\mathcal{E}\mathcal{E}} \equiv \frac{1}{m^2}\left\langle\mathbf{p}_{\epsilon}\cdot\mathbb{D}_{\epsilon}\cdot\mathbf{p}_{\epsilon}\right\rangle = \frac{2\mathcal{E}}{m}\left\langle D_{l\epsilon}\right\rangle = 2\mathcal{E}\frac{D_l}{m} + \mathcal{O}\left(\epsilon^2\right) \tag{23}$$

2.1.5 Calculation of $D_{gc}^{\mathcal{E}\mu}$

The expression for $D_{gc}^{\mathcal{E}\mu}$ reads

$$\begin{aligned}
D_{gc}^{\mathcal{E}\mu} \equiv& \frac{\Omega}{mB}\left\langle\mathbf{p}_{\epsilon}\cdot\mathbb{D}_{\epsilon}\cdot\frac{\partial\boldsymbol{\rho}_{\epsilon}}{\partial\varphi}\right\rangle \\
=& \frac{\Omega}{mB}\left\langle D_{l\epsilon}\frac{\partial\boldsymbol{\rho}_{\epsilon}}{\partial\varphi}\cdot\mathbf{p}_{\epsilon}\right\rangle
\end{aligned}$$

Similarly to the K_{gc}^{μ} calculation, we obtain

$$D_{gc}^{\mathcal{E}\mu} = \mu\frac{D_l}{m}\left(2 - \frac{v_{\parallel}}{\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right) + \mathcal{O}\left(\epsilon^2\right) \tag{24}$$

2.1.6 Calculation of $\mathbb{D}_{gc}^{\mathbf{X}\mathbf{X}}$

The expression for $\mathbb{D}_{gc}^{\mathbf{X}\mathbf{X}}$ reads

$$\begin{aligned}
\mathbb{D}_{gc}^{\mathbf{X}\mathbf{X}} \equiv& \left\langle\boldsymbol{\Delta}_{gc}^{\mathbf{X}T}\cdot\mathbb{D}_{\epsilon}\cdot\boldsymbol{\Delta}_{gc}^{\mathbf{X}}\right\rangle \\
=& \frac{1}{p^2}\left(D_l - D_t\right)\frac{\widehat{\mathsf{b}}}{m\Omega}\times\langle\mathbf{p}\,\mathbf{p}\rangle\times\frac{\widehat{\mathsf{b}}}{m\Omega} + \frac{1}{m^2\Omega^2}D_{t\epsilon}\left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}}\right) \\
=& \epsilon^2\frac{B}{m\Omega^2}\left(D_{gc}^{\mu} + \frac{D_t}{mB}\right)\left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}}\right) + \mathcal{O}\left(\epsilon^3\right)
\end{aligned}$$

where we used $\langle\mathbf{p}\,\mathbf{p}\rangle = m^2 v_{\parallel}^2\widehat{\mathsf{b}}\widehat{\mathsf{b}} + m\mu B\left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}}\right)$ and $D_{gc}^{\mu} \equiv \mu\left(D_l - D_t\right)/p^2$.

2.1.7 Calculation of $\mathbf{D}_{gc}^{\mathbf{X}\mathcal{E}}$

The expression for $\mathbf{D}_{gc}^{\mathbf{X}\mathcal{E}}$ reads

$$\begin{aligned}\mathbf{D}_{gc}^{\mathbf{X}\mathcal{E}} \equiv& \left\langle \boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbb{D}_\epsilon \cdot \boldsymbol{\Delta}_{gc}^{\mathcal{E}} \right\rangle \\ =& \frac{1}{m} \left\langle D_{l\epsilon} \boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbf{p}_\epsilon \right\rangle\end{aligned}$$

Similarly to $\mathbf{K}_{gc}^{\mathbf{X}}$, we therefore obtain:

$$\begin{aligned}\mathbf{D}_{gc}^{\mathbf{X}\mathcal{E}} =& - D_l \frac{\widehat{\mathsf{b}}}{m\Omega} \times \mathbf{v}_B - \frac{\mu B}{m\Omega^2} \boldsymbol{\nabla} D_l \cdot \left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}} \right) + \mathcal{O}\left(\epsilon^3\right) \\ =& \epsilon^2 \frac{\mu B}{m\Omega^2} \left[\nu \left(\boldsymbol{\nabla} \ln B + \frac{m v_\parallel^2}{\mu B} \widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \widehat{\mathsf{b}} \right) - \boldsymbol{\nabla} \nu \right] \cdot \left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}} \right) + \mathcal{O}\left(\epsilon^3\right)\end{aligned}$$

where $D_{gc}^{\mathcal{E}} \equiv D_l / m$.

2.1.8 Calculation of $\mathbf{D}_{gc}^{\mathbf{X}\mu}$

The expression for $\mathbf{D}_{gc}^{\mathbf{X}\mu}$ reads

$$\begin{aligned}\mathbf{D}_{gc}^{\mathbf{X}\mu} \equiv& \left\langle \boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbb{D}_\epsilon \cdot \boldsymbol{\Delta}_{gc}^{\mu} \right\rangle \\ =& \frac{1}{p^2} \left\langle \left(D_{l\epsilon} - D_{t\epsilon} \right) \left(\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbf{p}_\epsilon \right) \left(\boldsymbol{\Delta}_{gc}^{\mu} \cdot \mathbf{p}_\epsilon \right) \right\rangle + \left\langle D_{t\epsilon} \boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \boldsymbol{\Delta}_{gc}^{\mu} \right\rangle \\ =& \frac{1}{\mu} D_{gc}^{\mu} \left\langle \left(\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbf{p}_\epsilon \right) \left(\boldsymbol{\Delta}_{gc}^{\mu} \cdot \mathbf{p}_\epsilon \right) \right\rangle + D_t \left\langle \boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \boldsymbol{\Delta}_{gc}^{\mu} \right\rangle - 2 \boldsymbol{\nabla} D_{gc}^{\mu} \cdot \langle \boldsymbol{\rho}_0 \, \boldsymbol{\rho}_0 \rangle - \frac{1}{mB} \boldsymbol{\nabla} D_t \cdot \langle \boldsymbol{\rho}_0 \, \boldsymbol{\rho}_0 \rangle \\ =& - \frac{\mu B}{m\Omega^2} \left[D_{gc}^{\mu} \widehat{\mathsf{b}} \times \mathbf{v}_B + \boldsymbol{\nabla} \left(2 D^{\mu} + \frac{D_t}{mB} \right) \right] \cdot \left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}} \right) \\ =& \epsilon^2 \frac{\mu B}{m\Omega^2} \left[D_{gc}^{\mu} \left(\boldsymbol{\nabla} \ln B + \frac{m v_\parallel^2}{\mu B} \widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \widehat{\mathsf{b}} \right) - \boldsymbol{\nabla} \left(2 D^{\mu} + \frac{D_t}{mB} \right) \right] \cdot \left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}} \right) + \mathcal{O}\left(\epsilon^3\right)\end{aligned}$$

where we used the expressions found at (28), (29), (30), (31) and (32).

A Calculation of spatial projections

A.1 Preliminary calculations

Using (18), we have for a given vector or tensor $\mathbf{A}$:

$$\begin{aligned}\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbf{A} =& - \frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \boldsymbol{\nabla}^* \mathbf{X}_\epsilon \cdot \mathbf{A} - \mathbf{v}_{gc} \left(\frac{\partial \boldsymbol{\rho}_\epsilon}{\partial \mathcal{E}} \cdot \mathbf{A} \right) \\ =& - \frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \left(\mathbb{I} + \boldsymbol{\nabla} \boldsymbol{\rho}_\epsilon + \left[\mathbf{R} + \frac{1}{2} \left(\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}} \right) \widehat{\mathsf{b}} \right] \frac{\partial \boldsymbol{\rho}_\epsilon}{\partial \varphi} \right) \cdot \mathbf{A} - \mathbf{v}_{gc} \left(\frac{\partial \boldsymbol{\rho}_\epsilon}{\partial \mathcal{E}} \cdot \mathbf{A} \right)\end{aligned}$$

Using (8), we get at first order in ϵ:

$$\begin{aligned}\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbf{A} =& - \frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \left(\mathbb{I} - \frac{1}{2} \boldsymbol{\nabla} \ln B \, \boldsymbol{\rho}_0 - \left(\boldsymbol{\nabla} \widehat{\mathsf{b}} \cdot \boldsymbol{\rho}_0 \right) \widehat{\mathsf{b}} \right) \cdot \mathbf{A} \\ & - \mathbf{v}_{gc} \left(\frac{\partial \boldsymbol{\rho}_0}{\partial \mathcal{E}} \cdot \mathbf{A} \right) - v_\parallel \widehat{\mathsf{b}} \left(\frac{\partial \boldsymbol{\rho}_1}{\partial \mathcal{E}} \cdot \mathbf{A} \right) \\ =& - \epsilon \frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \left[\mathbf{A} - \epsilon \frac{1}{2} \boldsymbol{\nabla} \ln B \, \left(\boldsymbol{\rho}_0 \cdot \mathbf{A} \right) - \epsilon \left(\boldsymbol{\nabla} \widehat{\mathsf{b}} \cdot \boldsymbol{\rho}_0 \right) \left(\widehat{\mathsf{b}} \cdot \mathbf{A} \right) \right] \\ & - \epsilon^2 v_\parallel \widehat{\mathsf{b}} \left(\frac{\partial \boldsymbol{\rho}_1}{\partial \mathcal{E}} \cdot \mathbf{A} \right) + \mathcal{O}\left(\epsilon^3\right)\end{aligned} \tag{25}$$

where we used $\partial \boldsymbol{\rho}_0 / \partial \mathcal{E} = 0$. We can therefore deduce for a given symmetric tensor $\mathbb{A}$, at first order in ϵ:

$$\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbb{A} \cdot \boldsymbol{\Delta}_{gc}^{\mathbf{X}} = \boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \left(\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbb{A} \right)^T = \epsilon^2 \frac{\widehat{\mathsf{b}}}{m\Omega} \times \mathbb{A} \times \frac{\widehat{\mathsf{b}}}{m\Omega} + \mathcal{O}\left(\epsilon^3\right) \tag{26}$$

A.2 Calculation of $\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbb{I} \cdot \boldsymbol{\Delta}_{gc}^{\mathbf{X}}$

Using (26), we obtain at first order in ϵ:

$$\begin{aligned}\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbb{I} \cdot \boldsymbol{\Delta}_{gc}^{\mathbf{X}} =& \frac{\widehat{\mathsf{b}}}{m\Omega} \times \mathbb{I} \times \frac{\widehat{\mathsf{b}}}{m\Omega} \\ =& \epsilon^2 \frac{1}{m^2\Omega^2} \left(\mathbb{I} - \widehat{\mathsf{b}}\widehat{\mathsf{b}}\right) + \mathcal{O}\left(\epsilon^3\right)\end{aligned} \tag{27}$$

A.3 Calculation of $\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbf{p}_\epsilon$

Using (25) and (10), we obtain at first order in ϵ:

$$\begin{aligned}\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \mathbf{p}_\epsilon =& -\frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \left[\mathbf{p}_\epsilon - \frac{1}{2}\left(\boldsymbol{\rho}_0 \cdot \mathbf{p}\right) \boldsymbol{\nabla} \ln B - \left(\widehat{\mathsf{b}} \cdot \mathbf{p}\right)\left(\boldsymbol{\nabla}\widehat{\mathsf{b}} \cdot \boldsymbol{\rho}_0\right)\right] - v_\parallel \left(\frac{\partial \boldsymbol{\rho}_1}{\partial \mathcal{E}} \cdot \mathbf{p}\right) \widehat{\mathsf{b}} \\ =& -\epsilon \frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \left[\mathbf{p}_\epsilon - \epsilon m v_\parallel \left(\boldsymbol{\nabla}\widehat{\mathsf{b}} \cdot \boldsymbol{\rho}_0\right)\right] + \epsilon^2 v_\parallel \left(\frac{2\mu B}{\Omega}\frac{\partial \alpha}{\partial \mathcal{E}} + m v_\parallel \frac{\partial \gamma}{\partial \mathcal{E}}\right) \widehat{\mathsf{b}} + \mathcal{O}\left(\epsilon^3\right) \\ =& -\epsilon \frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \left[\mathbf{p}_\epsilon - \epsilon m v_\parallel \left(\boldsymbol{\nabla}\widehat{\mathsf{b}} \cdot \boldsymbol{\rho}_0\right)\right] + 2\epsilon^2 \left(\frac{\mu B}{m\Omega^2}\mathbb{A}_2 : \boldsymbol{\nabla}\widehat{\mathsf{b}} + 2\frac{v_\parallel}{\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla}\widehat{\mathsf{b}} \cdot \frac{\partial \boldsymbol{\rho}_0}{\partial \varphi}\right) \widehat{\mathsf{b}} + \mathcal{O}\left(\epsilon^3\right)\end{aligned} \tag{28}$$

A.4 Calculation of $\boldsymbol{\Delta}_{gc}^{\mu} \cdot \mathbf{p}_\epsilon$

The expression reads, using (10):

$$\begin{aligned}\boldsymbol{\Delta}_{gc}^{\mu} \cdot \mathbf{p}_\epsilon =& \frac{\Omega}{B}\frac{\partial \boldsymbol{\rho}_0}{\partial \varphi} \cdot \mathbf{p} + \frac{\Omega}{B}\frac{\partial \boldsymbol{\rho}_1}{\partial \varphi} \cdot \mathbf{p} + \frac{\Omega}{B}\frac{\partial \boldsymbol{\rho}_0}{\partial \varphi} \cdot \mathbf{p}_1 \\ =& 2\mu - \frac{\Omega}{B}\left[\frac{\mu B}{\Omega}\left(\beta + \frac{\partial \alpha}{\partial \varphi}\right) + m v_\parallel \frac{\partial \gamma}{\partial \varphi}\right] + \frac{\Omega}{B}\mathbf{v}_B \cdot \frac{\partial \boldsymbol{\rho}_0}{\partial \varphi} - \mu\left(\beta + \frac{\partial \alpha}{\partial \varphi} - \frac{v_\parallel}{2\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}}\right) \\ =& 2\mu + \epsilon \frac{m v_\parallel^2}{B}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla}\widehat{\mathsf{b}} \cdot \boldsymbol{\rho}_0 + \epsilon\mu\boldsymbol{\rho}_0 \cdot \boldsymbol{\nabla} \ln B - \epsilon\frac{\mu v_\parallel}{2\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}} + \epsilon\frac{\Omega}{B}\mathbf{v}_B \cdot \frac{\partial \boldsymbol{\rho}_0}{\partial \varphi} + \mathcal{O}\left(\epsilon^2\right)\end{aligned} \tag{29}$$

A.5 Calculation of $\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \boldsymbol{\Delta}_{gc}^{\mu}$

The expression reads, using (25) and (20):

$$\begin{aligned}\boldsymbol{\Delta}_{gc}^{\mathbf{X}T} \cdot \boldsymbol{\Delta}_{gc}^{\mu} =& -\frac{\Omega}{B}\frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*} \times \frac{\partial \boldsymbol{\rho}_\epsilon}{\partial \varphi} - \frac{\Omega v_\parallel}{B}\left(\frac{\partial \boldsymbol{\rho}_1}{\partial \mathcal{E}} \cdot \frac{\partial \boldsymbol{\rho}_0}{\partial \varphi}\right) \widehat{\mathsf{b}} \\ =& \frac{1}{mB}\left(1 - \frac{v_\parallel}{\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}}\right)\left[\left(\frac{\partial \alpha}{\partial \varphi} + \beta - 1\right)\boldsymbol{\rho}_0 - \left(\frac{\partial \beta}{\partial \varphi} - \alpha\right)\frac{\partial \boldsymbol{\rho}_0}{\partial \varphi}\right] + 2\frac{\mu v_\parallel}{\Omega}\frac{\partial \alpha}{\partial \mathcal{E}}\widehat{\mathsf{b}} \\ =& \epsilon\frac{1}{mB}\boldsymbol{\rho}_0 + \epsilon^2\frac{1}{mB}\left[\frac{v_\parallel}{2\Omega}\mathbb{A}_1 : \boldsymbol{\nabla}\widehat{\mathsf{b}} - \frac{1}{2}\boldsymbol{\rho}_0 \cdot \boldsymbol{\nabla} \ln B - \frac{v_\parallel}{2\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}}\right]\boldsymbol{\rho}_0 \\ &+ \epsilon^2\frac{1}{mB}\left[\frac{v_\parallel}{\Omega}\frac{\partial \mathbb{A}_1}{\partial \varphi} : \boldsymbol{\nabla}\widehat{\mathsf{b}} - + \frac{1}{2}\frac{\partial \boldsymbol{\rho}_0}{\partial \varphi} \cdot \boldsymbol{\nabla} \ln B + \frac{v_\parallel}{\Omega}\mathbb{A}_1 : \boldsymbol{\nabla}\widehat{\mathsf{b}} + \frac{v_\parallel}{\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}}\right]\frac{\partial \boldsymbol{\rho}_0}{\partial \varphi} \\ &+ 2\epsilon^2\left[\frac{\mu}{m\Omega^2}\mathbb{A}_2 : \boldsymbol{\nabla}\widehat{\mathsf{b}} + \frac{v_\parallel}{B\Omega^2}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla}\widehat{\mathsf{b}} \cdot \frac{\partial \boldsymbol{\rho}_0}{\partial \varphi}\right]\widehat{\mathsf{b}} + \mathcal{O}\left(\epsilon^3\right)\end{aligned} \tag{30}$$

where we used $\Omega_\parallel^* = \Omega\left(1 + \epsilon\frac{v_\parallel}{\Omega}\widehat{\mathsf{b}} \cdot \boldsymbol{\nabla} \times \widehat{\mathsf{b}}\right)$.

A.6 Calculation of $\left\langle \left(\boldsymbol{\Delta}_{gc}^{\mathbf{X}T}\cdot\mathbf{p}_\epsilon\right)\left(\boldsymbol{\Delta}_{gc}^{\mu}\cdot\mathbf{p}_\epsilon\right)\right\rangle$

Using (28) and (29), we have:

$$\begin{aligned}\left\langle \left(\boldsymbol{\Delta}_{gc}^{\mathbf{X}T}\cdot\mathbf{p}_\epsilon\right)\left(\boldsymbol{\Delta}_{gc}^{\mu}\cdot\mathbf{p}_\epsilon\right)\right\rangle =& -2\mu\frac{\widehat{\mathsf{b}}}{m\Omega_\parallel^*}\times\left[\langle\mathbf{p}_\epsilon\rangle-\epsilon m v_\parallel\left(\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\langle\boldsymbol{\rho}_0\rangle\right)\right]+4\mu\left(\frac{\mu B}{m\Omega^2}\langle\mathbb{A}_2\rangle:\boldsymbol{\nabla}\widehat{\mathsf{b}}+2\frac{v_\parallel}{\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle\right)\widehat{\mathsf{b}}\\ &-\frac{mv_\parallel^2}{B}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\langle\boldsymbol{\rho}_0\,\boldsymbol{\rho}_0\rangle-\mu\langle\boldsymbol{\rho}_0\,\boldsymbol{\rho}_0\rangle\cdot\boldsymbol{\nabla}\ln B+\langle\boldsymbol{\rho}_0\rangle\frac{\mu v_\parallel}{2\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}-\frac{\Omega}{B}\mathbf{v}_B\cdot\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\boldsymbol{\rho}_0\right\rangle\\ =&-2\mu\frac{\widehat{\mathsf{b}}}{\Omega}\times\mathbf{v}_B-\mu\frac{v_\parallel^2}{\Omega^2}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\left(\mathbb{I}-\widehat{\mathsf{b}}\widehat{\mathsf{b}}\right)-\mu\frac{\mu B}{m\Omega^2}\boldsymbol{\nabla}\ln B\cdot\left(\mathbb{I}-\widehat{\mathsf{b}}\widehat{\mathsf{b}}\right)\\ =&\epsilon^2\frac{\mu^2 B}{m\Omega^2}\left[\boldsymbol{\nabla}\ln B+\frac{mv_\parallel^2}{\mu B}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\right]\cdot\left(\mathbb{I}-\widehat{\mathsf{b}}\widehat{\mathsf{b}}\right)+\mathcal{O}\left(\epsilon^3\right)\end{aligned}\tag{31}$$

since $\langle\boldsymbol{\rho}_0\rangle=\left\langle\boldsymbol{\rho}_0\,\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle=\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle=\langle\mathbb{A}_2\rangle=0$ and $\langle\boldsymbol{\rho}_0\,\boldsymbol{\rho}_0\rangle=\left(\mu B/m\Omega^2\right)\left(\mathbb{I}-\widehat{\mathsf{b}}\widehat{\mathsf{b}}\right)$.

A.7 Calculation of $\left\langle\boldsymbol{\Delta}_{gc}^{\mathbf{X}T}\cdot\boldsymbol{\Delta}_{gc}^{\mu}\right\rangle$

Using (30), the expression reads:

$$\begin{aligned}\left\langle\boldsymbol{\Delta}_{gc}^{\mathbf{X}T}\cdot\boldsymbol{\Delta}_{gc}^{\mu}\right\rangle=&\frac{1}{mB}\left[\langle\boldsymbol{\rho}_0\rangle+\frac{v_\parallel}{2\Omega}\langle\boldsymbol{\rho}_0\,\mathbb{A}_1\rangle:\boldsymbol{\nabla}\widehat{\mathsf{b}}-\frac{1}{2}\langle\boldsymbol{\rho}_0\,\boldsymbol{\rho}_0\rangle\cdot\boldsymbol{\nabla}\ln B-\frac{v_\parallel}{2\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\langle\boldsymbol{\rho}_0\rangle\right.\\ &+\frac{v_\parallel}{\Omega}\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\frac{\partial\mathbb{A}_1}{\partial\varphi}\right\rangle:\boldsymbol{\nabla}\widehat{\mathsf{b}}+\frac{1}{2}\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle\cdot\boldsymbol{\nabla}\ln B\\ &\left.+\frac{v_\parallel}{\Omega}\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\mathbb{A}_1\right\rangle:\boldsymbol{\nabla}\widehat{\mathsf{b}}+\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle\frac{v_\parallel}{\Omega}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\times\widehat{\mathsf{b}}\right]\\ &+2\left[\frac{\mu}{m\Omega^2}\langle\mathbb{A}_2\rangle:\boldsymbol{\nabla}\widehat{\mathsf{b}}+\frac{v_\parallel}{B\Omega^2}\widehat{\mathsf{b}}\cdot\boldsymbol{\nabla}\widehat{\mathsf{b}}\cdot\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle\right]\widehat{\mathsf{b}}\\ =&\mathcal{O}\left(\epsilon^3\right)\end{aligned}\tag{32}$$

since $\langle\boldsymbol{\rho}_0\rangle=\langle\boldsymbol{\rho}_0\,\mathbb{A}_1\rangle=\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\frac{\partial\mathbb{A}_1}{\partial\varphi}\right\rangle=\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\mathbb{A}_1\right\rangle=\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle=\langle\mathbb{A}_2\rangle=0$ and $\langle\boldsymbol{\rho}_0\,\boldsymbol{\rho}_0\rangle=\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\,\frac{\partial\boldsymbol{\rho}_0}{\partial\varphi}\right\rangle=\left(\mu B/m\Omega^2\right)\left(\mathbb{I}-\widehat{\mathsf{b}}\widehat{\mathsf{b}}\right)$.

References

[1] A. J. Brizard, A guiding-center Fokker-Planck collision operator for nonuniform magnetic fields, *Phys. Plasmas*, **11**(9):4429–4438, 2004.